

Customer Churn Analytics

Credit card and e-commerce customers

961731 Business Data Analytics · Chiang Mai University

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13 September 2026

Credit card: dataset architecture

- Kaggle uploader: sakshigoyal7.
- Source: 10,127 records × 23 columns. Churn means Attrited Customer = 1.
- Source churn share: 16.1%. Analytical cohort: 10,127 records.
- The label horizon and feature cut-off dates are unspecified. Results describe a historical snapshot.

Field	Meaning
Attrition	Observed status
Trans. count	Card activity
Inactive	Idle months
Utilization	Used credit share
Relations	Products held

Source: Kaggle: sakshigoyal7; lecture computation

Credit card: five customer records

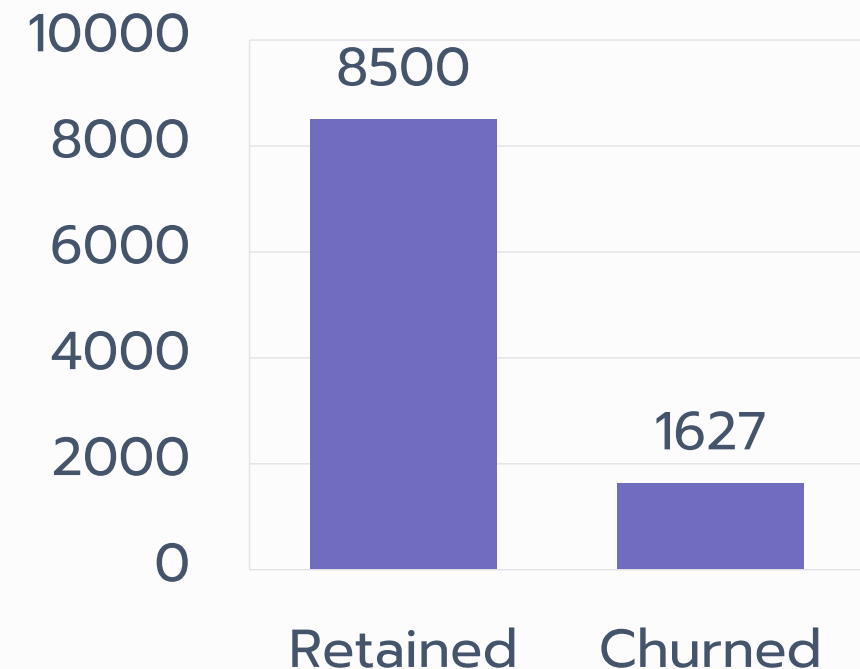
- The table shows five genuine source rows.
y = 1 marks observed churn.
- Age = years; Txns = transaction count; Idle = inactive months; Rel. = products; Util. = used credit share.
- Identifiers stay outside the model.
Descriptive rows cannot establish a future prediction window.
- A data dictionary should preserve units and time windows. Unspecified units remain explicit.

y	Age	Txns	Idle	Rel.	Util.
0	45	42	1	5	0.06
0	49	33	1	6	0.10
0	51	20	1	4	0
0	40	20	4	3	0.76
0	40	28	1	5	0

Credit card: baseline statistical integrity

- Analytical churn: $1,627/10,127 = 16.1\%$. Majority-class accuracy alone is 83.9%.
- Source missing cells: 0. Keep Unknown categories and fit imputers within training folds.
- The source contains no missing cells, but Unknown appears in demographic categories.
- EDA uses training records. Missing customer timestamps prevent a claim of future stability.

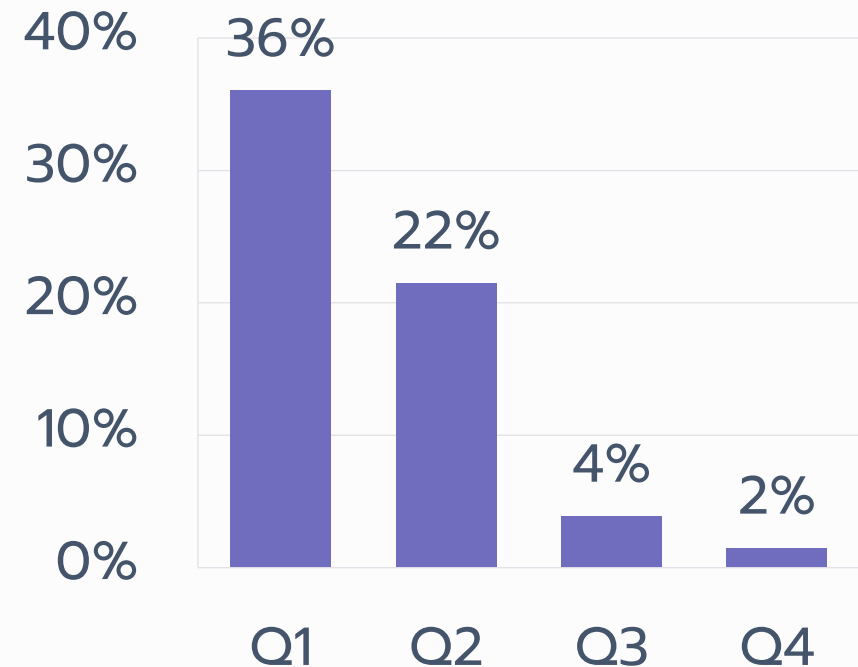
Analytical class distribution



Credit card: transaction activity

- Highest observed group: (9.999, 45.0], 36.1% churn (n = 1,564).
- Lowest observed group: (81.0, 134.0], 1.5% churn (n = 1,398).
- Low card usage can motivate a reactivation experiment.
- These are training-set associations. Group mix, label timing and confounding can explain part of the gap.

Churn share, training set



Credit card: inactivity

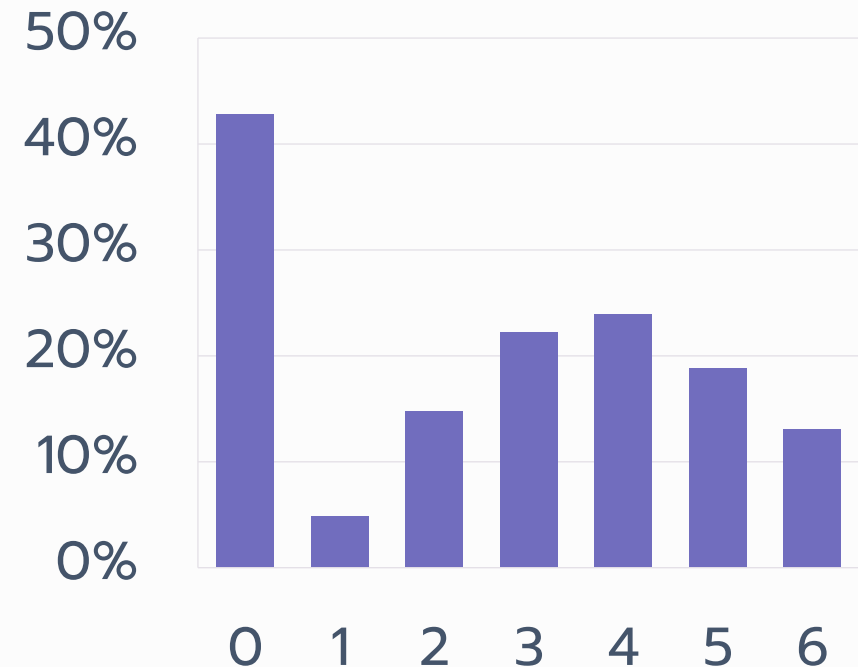
Highest observed group: 0, 42.9% churn (n = 21).

Lowest observed group: 1, 4.9% churn (n = 1,277).

Idle periods can trigger a service check before incentives.

The zero-inactivity group has only 21 training records. Its high rate is too uncertain to define a production rule.

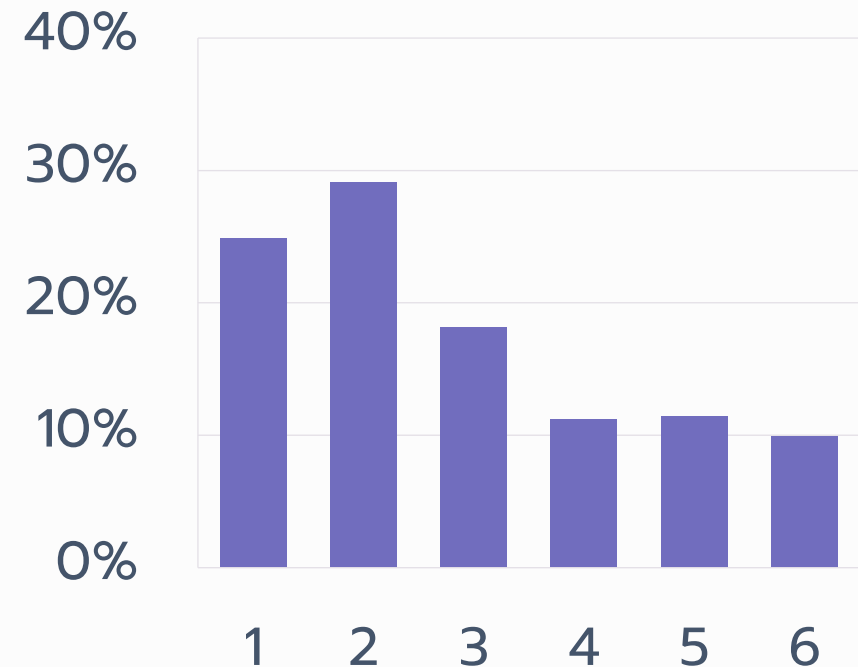
Churn share, training set



Credit card: relationship depth

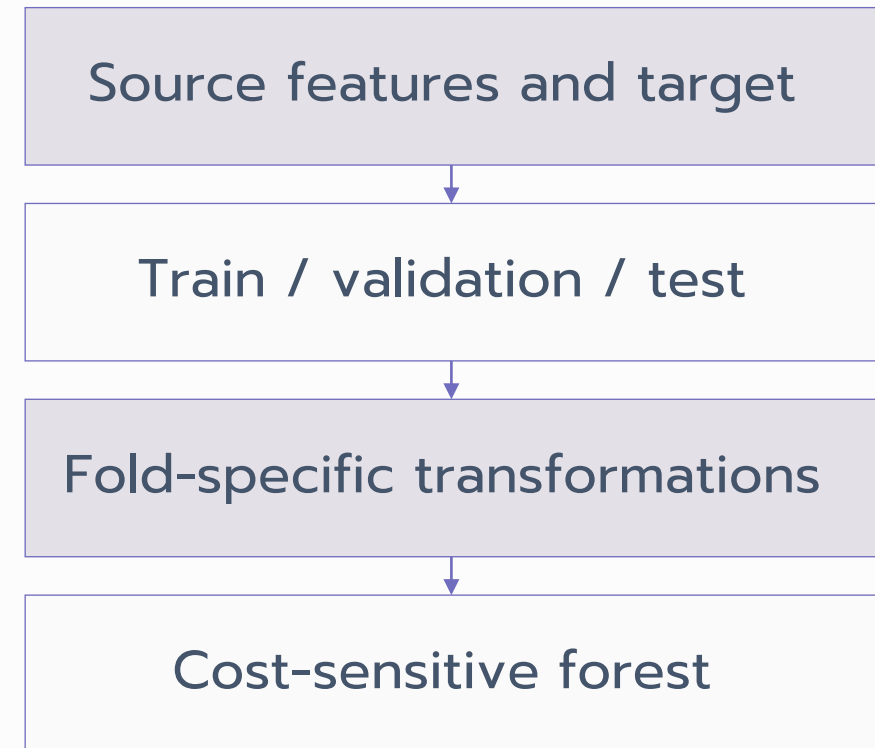
- Highest observed group: 2, 29.1% churn (n = 772).
- Lowest observed group: 6, 10.0% churn (n = 1,115).
- Product depth can guide differentiated account reviews.
- These are training-set associations. Group mix, label timing and confounding can explain part of the gap.

Churn share, training set



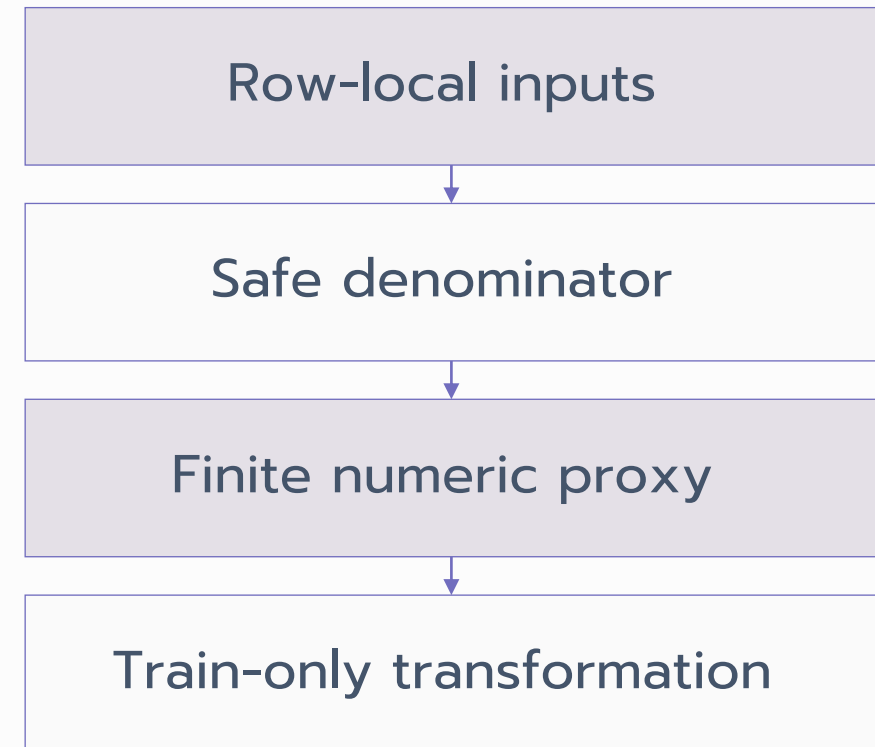
Credit card: feature preprocessing

- Drop CLIENTNUM and both Naive Bayes output columns. The latter encode information tied to the target.
- Numeric branch: median imputation, then StandardScaler. Categorical branch: mode imputation, then OneHotEncoder.
- ColumnTransformer fits inside each CV fold. Unseen categories use `handle_unknown='ignore'`.
- Scaling supports a reusable pipeline. Trees do not require standardized units.



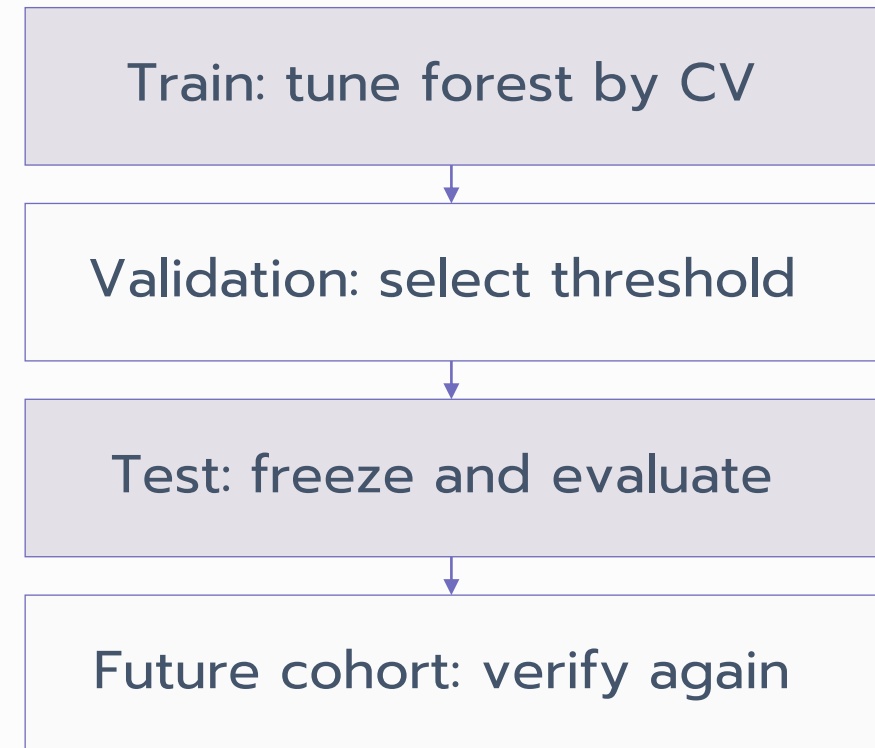
Credit card: domain feature engineering

- Average ticket = transaction amount / $\max(\text{transaction count}, 1)$.
- Contact ratio = annual contacts / $\max(\text{relationships}, 1)$.
- Ratios add business structure but share information with their parent features.
- A ratio is eligible only when every input is known at the scoring date.



Credit card: cost-sensitive random forest

- A forest averages churn probabilities across decision trees. Each tree learns split rules from a bootstrap sample.
- Balanced class weights increase the influence of the less frequent churn class during fitting.
- Data partition: 6,076 train, 2,025 validation and 2,026 test records.
- Training chooses the model. Validation chooses an action threshold. The test partition assesses the frozen decision rule.



Credit card: hyperparameter optimization

- GridSearchCV selects by average precision across three stratified training folds.
- Depth controls tree complexity. Leaf size controls the smallest terminal customer group.
- The grid tests 12 candidates with 150 trees and max_features in {sqrt, 0.6}.
- Every fold refits preprocessing. CV selects the model; the separate test set estimates final performance.

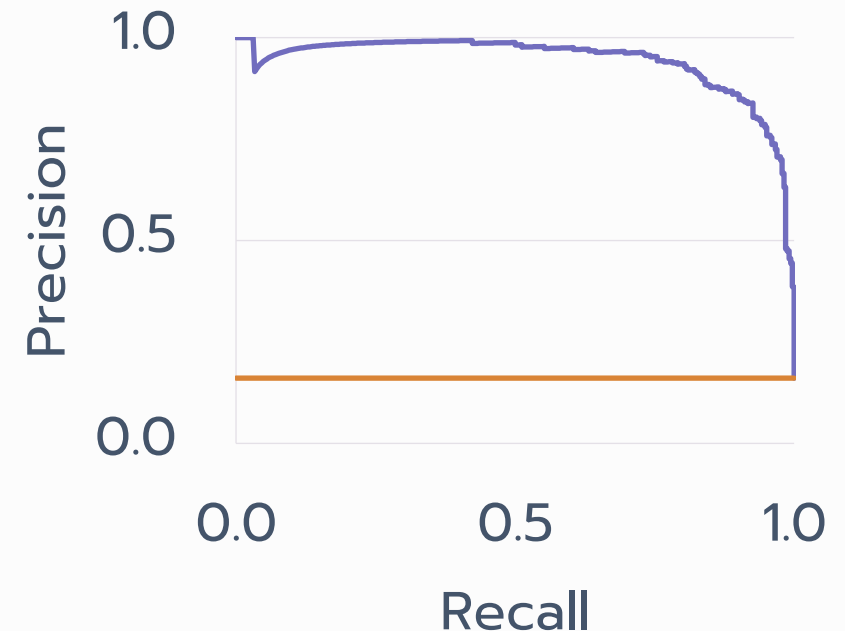
```
grid = {  
    'model__max_depth':  
        [5, 10, None],  
    'model__min_samples_leaf':  
        [2, 8],  
    'model__max_features':  
        ['sqrt', 0.6]  
}  
search = GridSearchCV(  
    pipeline, grid, cv=cv,  
    scoring='average_precision',  
    n_jobs=2  
)  
search.fit(X_train, y_train)
```

Source: [scikit-learn, GridSearchCV](#)

Credit card: ranking performance

- Test PR-AUC (average precision): 0.943.
Test ROC-AUC: 0.988.
- Precision measures how many flagged customers churned. Recall measures how many churners were captured.
- Random-ranking precision is the test churn share, 16.1%.
- Average precision weights precision by recall increments. It differs from trapezoidal PR area.

Precision-recall, held-out test



Credit card: decision costs

- Validation selected threshold 0.261. Test recall is 95.1% and precision is 76.9%.
- Teaching cost assumption: missed churn costs 5 units; unnecessary contact costs 1 unit.
- Test loss = $5 \times 16 + 93 = 173$ cost units.
- Replace these illustrative costs with verified economics. Detection does not establish that an offer will prevent churn.

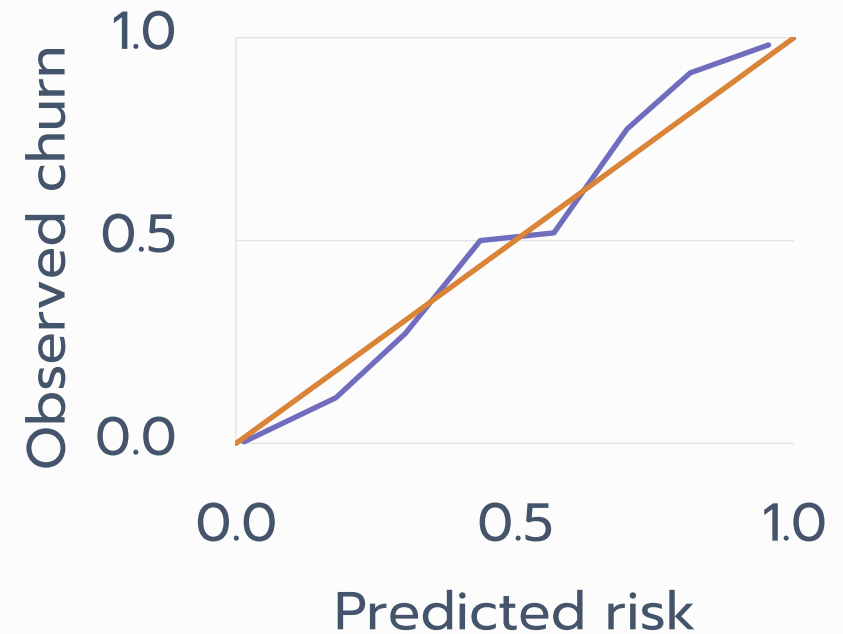
Actual / flag	No	Yes
Retained	1607	93
Churned	16	310

False positive: +1 cost; false negative: +5 cost

Credit card: probability calibration

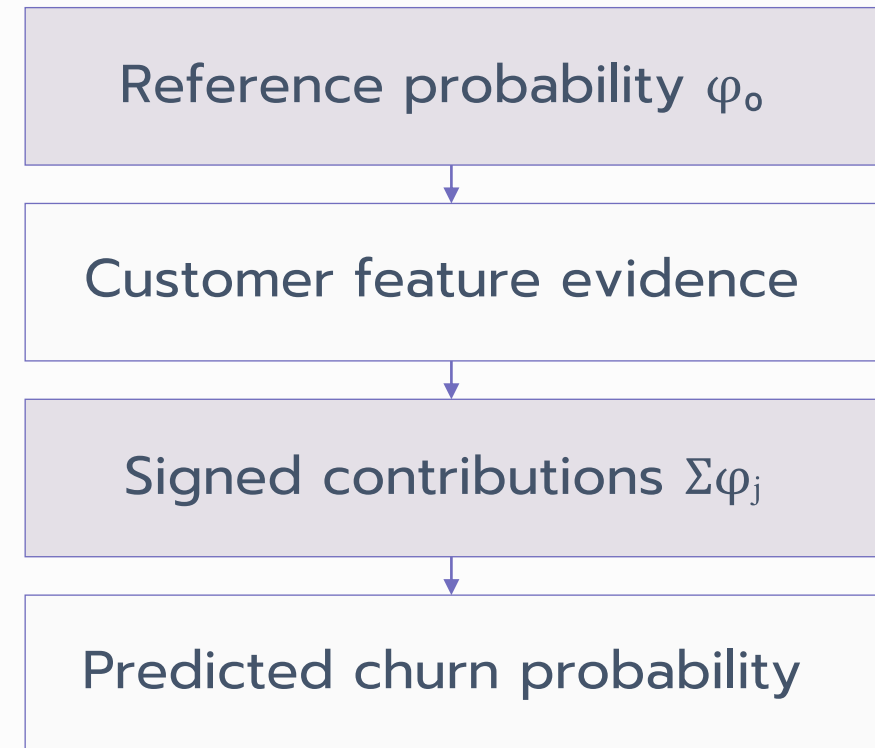
- Test Brier score: 0.0301. Lower values indicate lower mean squared probability error.
- Each bin compares mean predicted risk with observed churn share.
- Above the diagonal means underprediction in that risk region; below means overprediction.
- This is an evaluation. Later recalibration needs training data and fresh holdout validation.

Reliability diagram, test



Credit card: game-theoretic explanation

- SHAP treats model inputs as players in a cooperative prediction game.
- The explained value is this forest's churn probability for one customer, relative to its training-path reference.
- Each feature receives a signed contribution. The contributions and baseline sum to the prediction.
- A global summary aggregates many local explanations. It describes how the fitted model uses information.



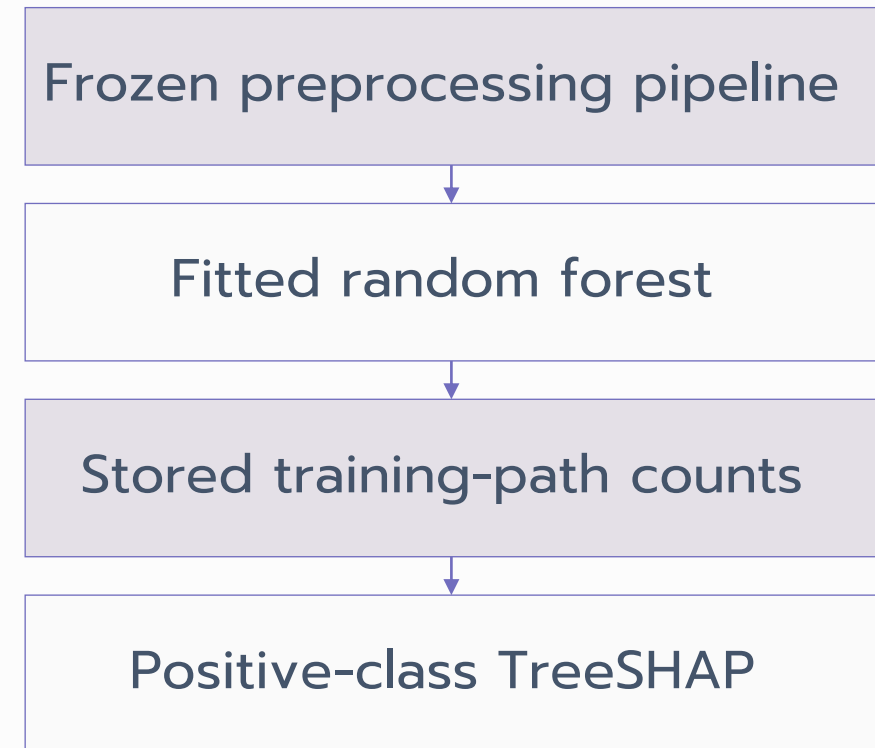
Credit card: the Shapley allocation

- A feature's value is its average marginal contribution across all coalition positions.
- For coalition S without j , the increment is $v(S \cup \{j\}) - v(S)$.
- The Shapley weight is $|S|!(M - |S| - 1)! / M!$, where M is the number of transformed features.
- Local accuracy requires $f(x) = \varphi_0 + \sum_j \varphi_j$. Positive values raise the model output relative to the baseline.

Property	Meaning
Efficiency	Contributions add
Symmetry	Equal role, credit
Null player	No effect, zero
Reference	Baseline matters

Credit card: TreeSHAP computation

- TreeSHAP exploits tree paths to avoid enumerating all 2^M feature coalitions.
- The fitted forest is the baseline model and the explanation target. SHAP does not train a stronger churn model.
- KernelSHAP is model-agnostic and samples coalitions. It is useful when a tree-specific engine is unavailable.
- Path-dependent TreeSHAP uses stored training frequencies. Raw output for this forest is class probability.



Credit card: explanation boundaries

- Correlated inputs and derived ratios can share credit. Rankings depend on the model reference.
- Training-path attribution does not identify causal effects or the true conditional distribution.
- Grouped dummy columns preserve the total. They aggregate the original game for display.
- Positive SHAP means higher model risk. Retention impact requires a controlled study.

Evidence	Interpretation
High risk	Prioritize review
Positive φ	Raises prediction
Mean $ \varphi $	Model reliance
Randomized lift	Action effect

Source: SHAP, TreeExplainer documentation; Lundberg & Lee (2017), SHAP

Credit card: advanced engine architecture

- Transform test rows with the frozen ColumnTransformer. The model stores the training-path reference.
- For a scikit-learn random forest, `model_output='raw'` returns class probabilities, including churn class 1.
- Resolve the class axis across SHAP versions before constructing a `shap.Explanation`.
- Assert that the baseline plus feature contributions reproduces each customer's predicted risk.

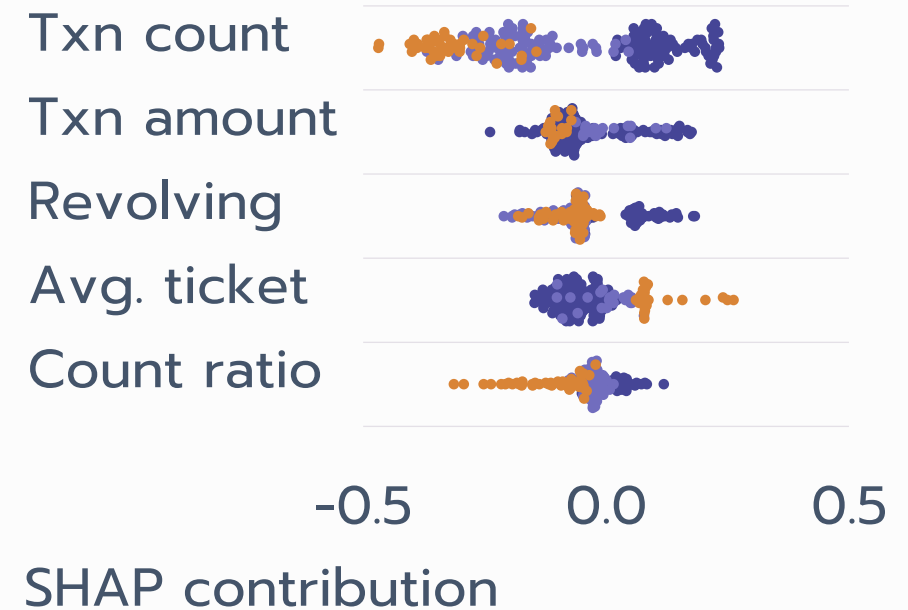
```
engine = shap.TreeExplainer(
    forest, model_output='raw',
    feature_perturbation=
        'tree_path_dependent'
)
raw = engine.shap_values(Z)
phi = raw[:, :, 1]
base = engine.expected_value[1]
np.testing.assert_allclose(
    base + phi.sum(1), p,
    atol=1e-5, rtol=0
)
```

Source: SHAP, TreeExplainer documentation

Credit card: global SHAP beeswarm

- Largest mean absolute attribution: Txn count, 18.8 percentage points.
- Each dot is one explained test customer. Its horizontal position is that feature's contribution to predicted churn.
- Blue marks lower numeric values; orange marks higher ones. Categorical and missing values use neutral purple.
- Feature order summarizes model reliance. It does not rank the profit from changing those features.

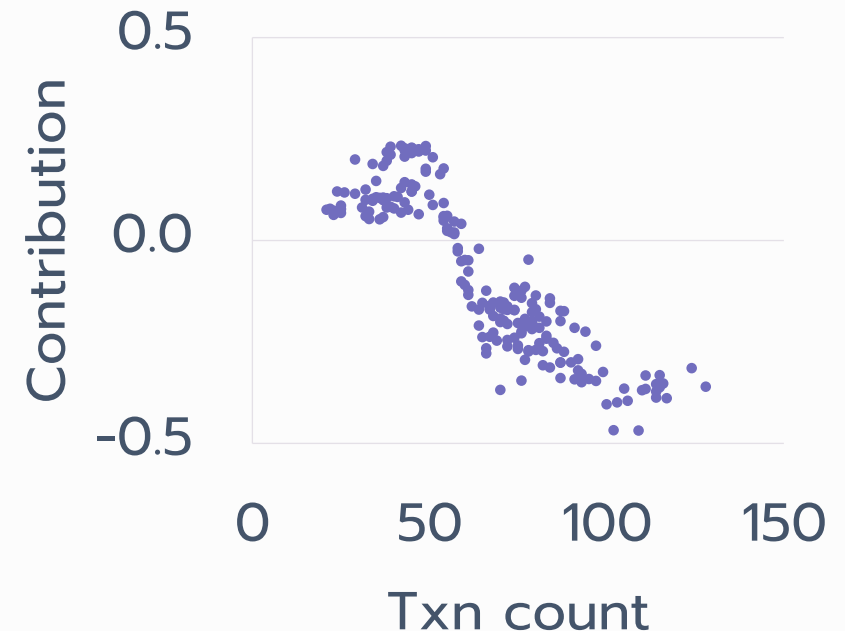
TreeSHAP, top five features



Credit card: nonlinear attribution pattern

- Horizontal: txn count. Vertical: its grouped contribution to predicted churn.
- A sign change marks a contribution crossing zero relative to the model reference.
- A bend suggests a candidate trigger. Its location needs stability testing on future data.
- Vertical spread reflects other inputs and interactions. Sparse tails need caution.

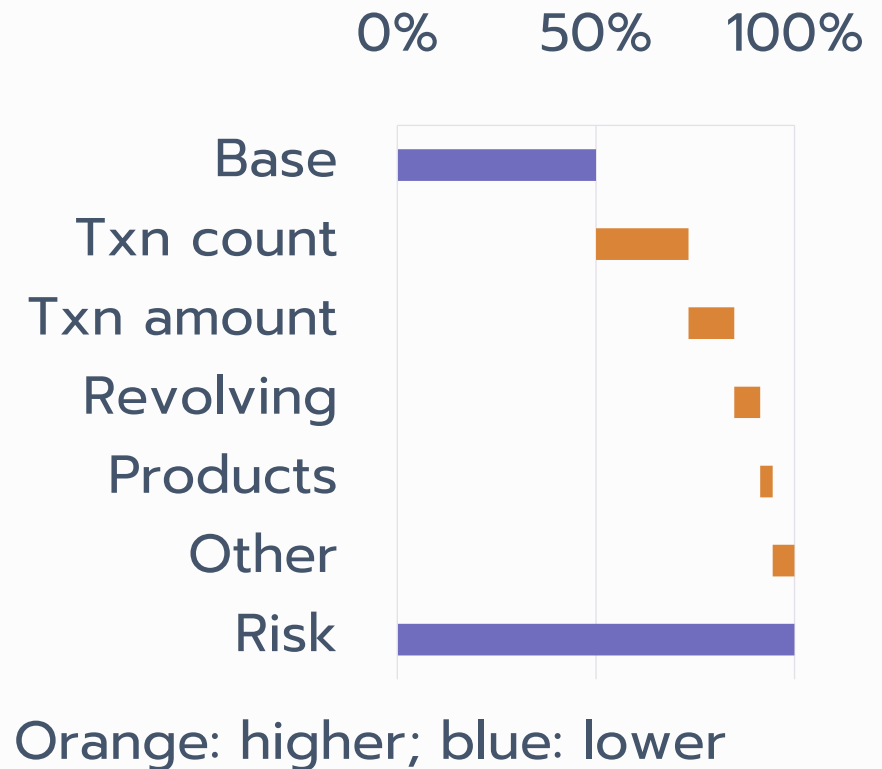
SHAP dependence, test sample



Credit card: one high-risk customer

- Model reference: 50.0%. This customer's predicted risk: 100.0%.
- The waterfall starts at the reference probability and adds each signed contribution.
- The four largest effects are visible. Other features aggregate the remaining contributions exactly.
- The customer is selected by predicted risk, without using the observed label to choose the example.

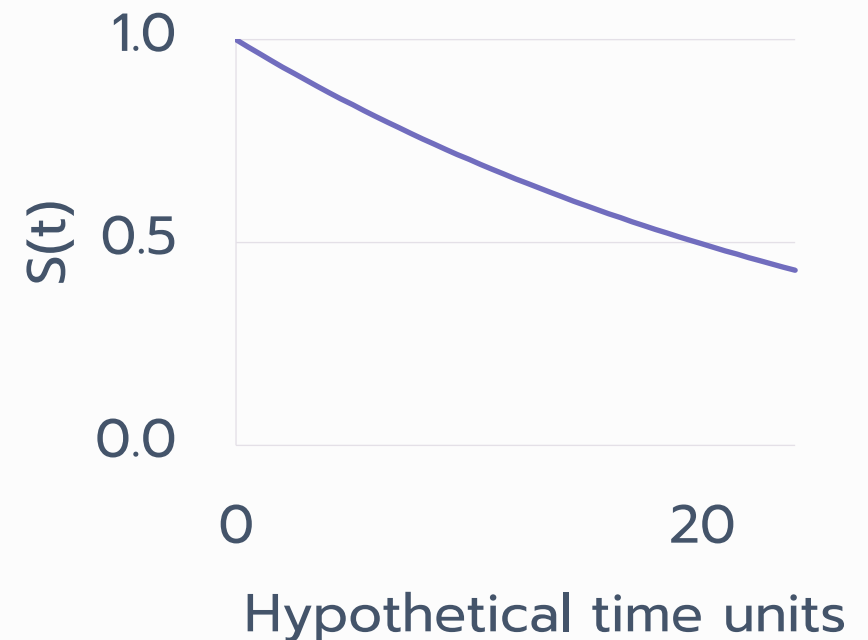
Local probability waterfall



Credit card: survival and hazard over time

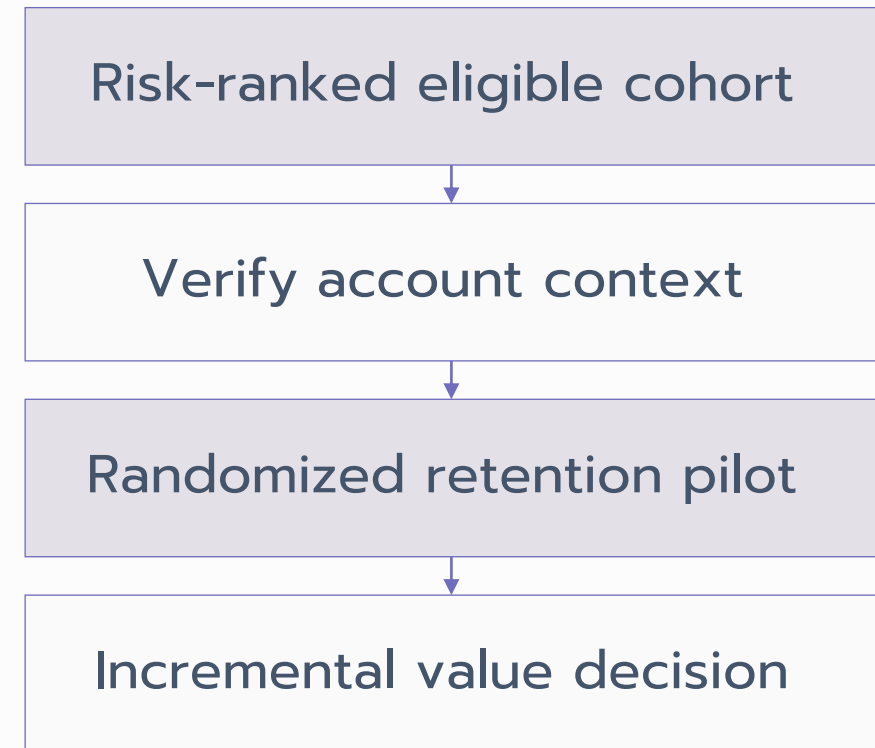
- Survival is $S(t) = P(T > t)$. Hazard is the event rate among customers still at risk.
- Event dates, censoring dates and an observation origin are absent. Empirical survival is not identified.
- Hypothetical only: hazard $\lambda = 0.035$ per time unit; $S(t) = \exp(-\lambda t)$.
- A longitudinal extension needs join date, event or censoring time and features known before each interval.

Hypothetical survival only



Credit card: executive retention playbook

- Observed evidence: test AP 0.943; the largest model attribution is txn count.
- Priority hypothesis: review disengaged card users and service-contact friction.
- Randomize eligible accounts to treatment and control. Report incremental retained margin after contact and incentive costs.
- Launch gate: verified timing, independent future validation and a positive incremental value estimate. Pause if any gate fails.



E-commerce: dataset architecture

- Kaggle uploader: ankitverma2010. The original organization is not identified.
- Source: 5,630 records × 20 columns. Churn means Churn = 1.
- Source churn share: 16.8%. Analytical cohort: 5,074 records.
- The label horizon and feature cut-off dates are unspecified. Results describe a historical snapshot.

Field	Meaning
Churn	Observed status
Tenure	Units unspecified
Complain	Complaint flag
Last order	Recency in days
Cashback	Monthly average

Source: Kaggle: ankitverma2010; lecture computation

E-commerce: five customer records

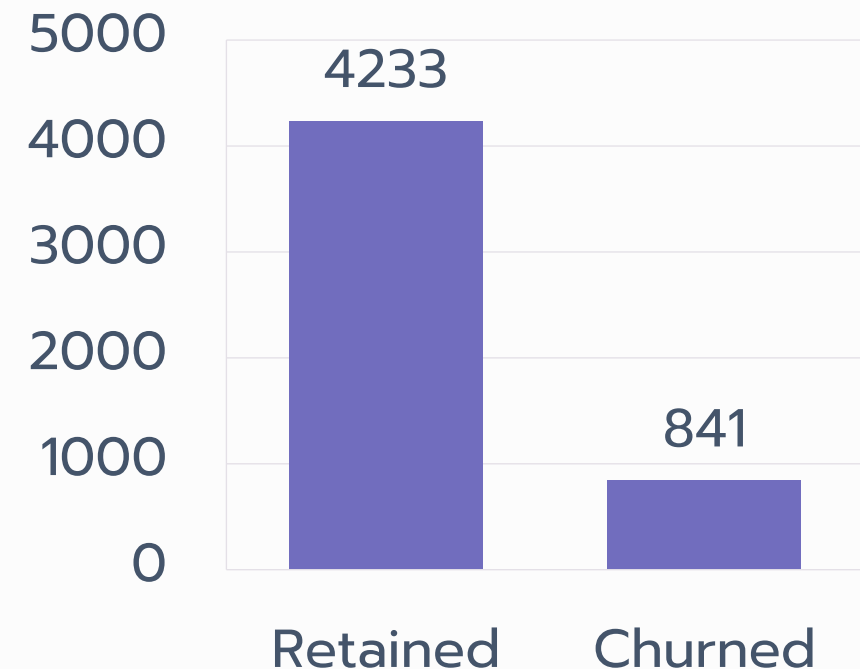
- The table shows five genuine source rows.
y = 1 marks observed churn.
- Ten. = tenure; Issue = complaint flag; Ord. = orders; Days = last-order recency; Cash = cashback.
- Identifiers stay outside the model.
Descriptive rows cannot establish a future prediction window.
- A data dictionary should preserve units and time windows. Unspecified units remain explicit.

y	Ten.	Issue	Ord.	Days	Cash
1	4	1	1	5	159.93
1	NA	1	1	0	120.90
1	NA	1	1	3	120.28
1	0	0	1	3	134.07
1	0	0	1	3	129.60

E-commerce: baseline statistical integrity

- Analytical churn: $841/5,074 = 16.6\%$.
Majority-class accuracy alone is 83.4%.
- Source missing cells: 1,856. Keep Unknown categories and fit imputers within training folds.
- Remove 556 repeated profiles before splitting. This changes the cohort denominator.
- EDA uses training records. Missing customer timestamps prevent a claim of future stability.

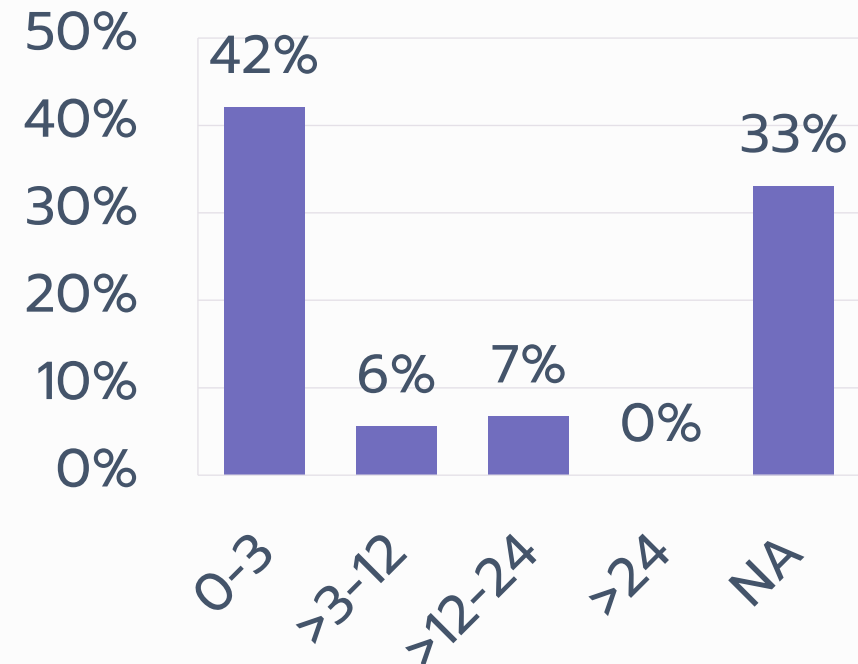
Analytical class distribution



E-commerce: tenure

- Highest observed group: 0-3, 42.1% churn (n = 834).
- Lowest observed group: >24, 0.0% churn (n = 237).
- Early-tenure customers may benefit from onboarding support.
- These are training-set associations. Group mix, label timing and confounding can explain part of the gap.

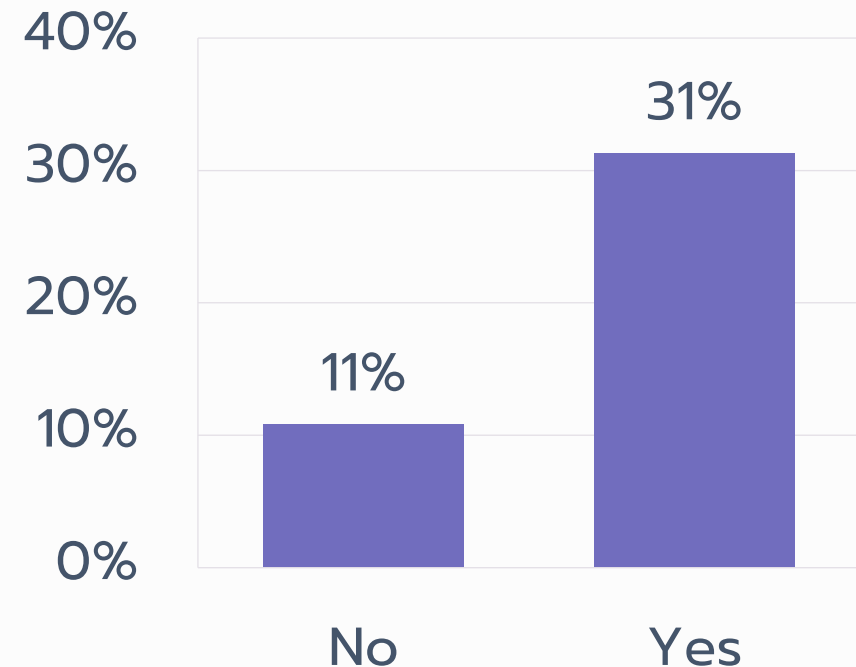
Churn share, training set



E-commerce: complaints

- Highest observed group: Complaint, 31.3% churn (n = 860).
- Lowest observed group: No complaint, 10.8% churn (n = 2,184).
- Complaint handling is a testable service intervention.
- These are training-set associations. Group mix, label timing and confounding can explain part of the gap.

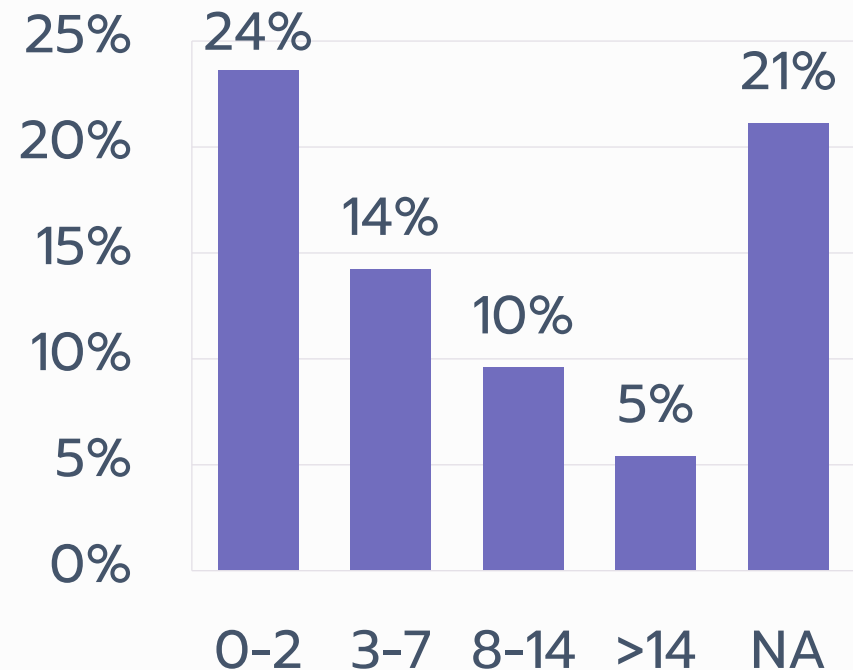
Churn share, training set



E-commerce: order recency

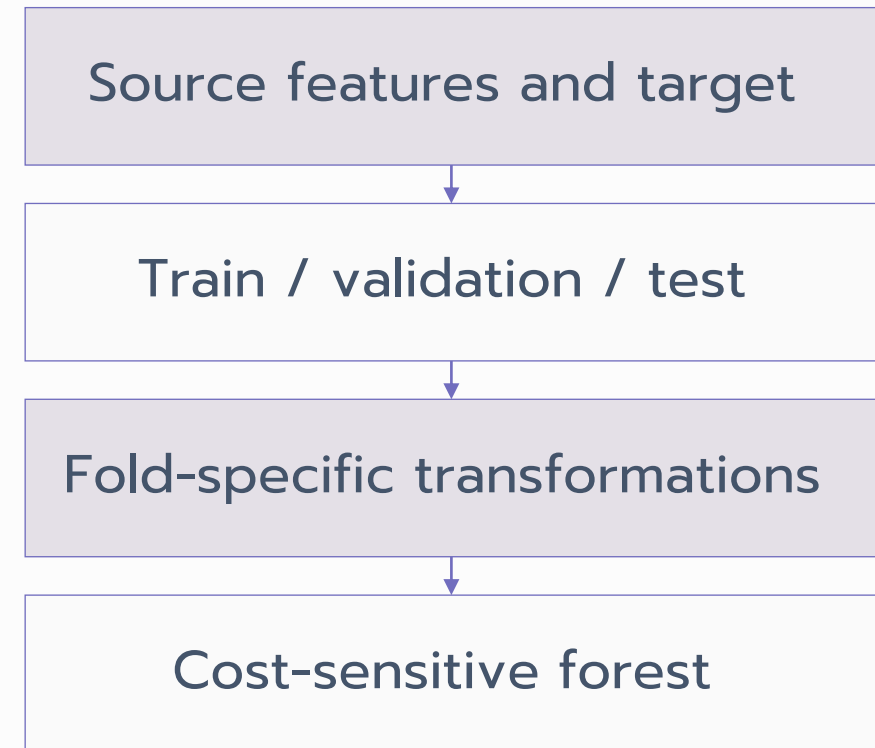
- Highest observed group: 0-2 days, 23.6% churn (n = 1,011).
- Lowest observed group: >14 days, 5.4% churn (n = 37).
- Recency can help time a re-engagement experiment.
- These are training-set associations. Group mix, label timing and confounding can explain part of the gap.

Churn share, training set



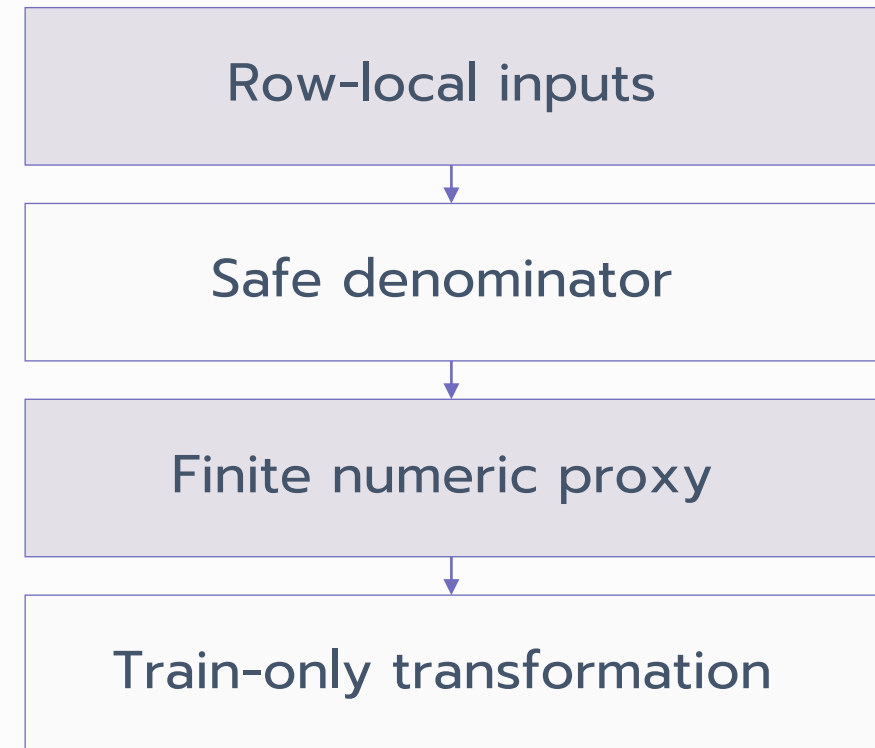
E-commerce: feature preprocessing

- Drop CustomerID and remove exact repeated profiles. Keep complaint and satisfaction timing under review.
- Numeric branch: median imputation, then StandardScaler. Categorical branch: mode imputation, then OneHotEncoder.
- ColumnTransformer fits inside each CV fold. Unseen categories use `handle_unknown='ignore'`.
- Scaling supports a reusable pipeline. Trees do not require standardized units.



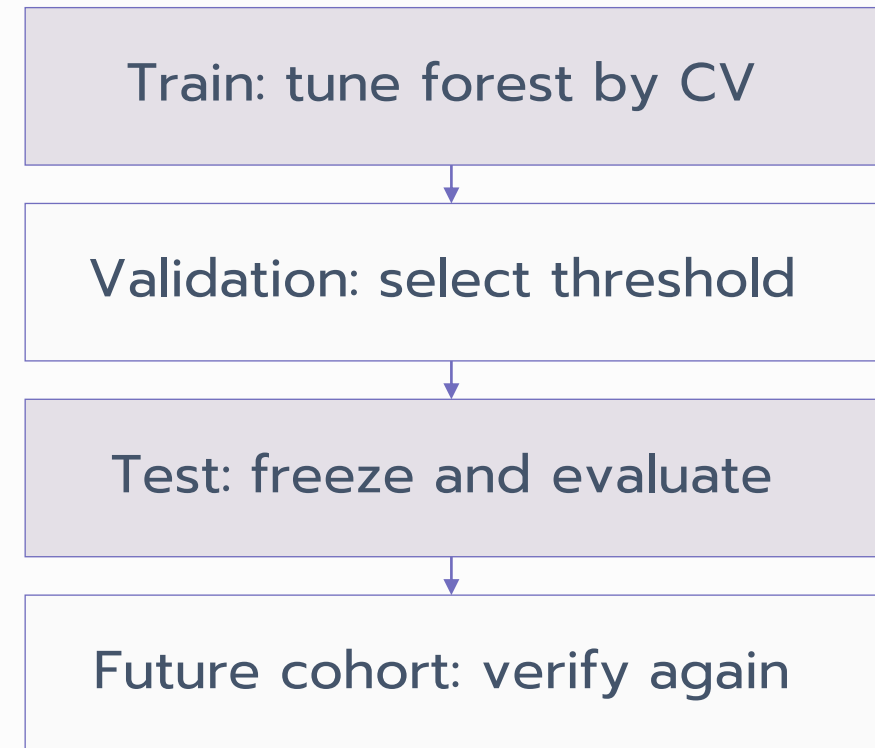
E-commerce: domain feature engineering

- Cashback ratio = monthly average cashback / max(order count, 1).
- Order intensity = order count / (tenure + 1). The +1 smooths zero tenure.
- These are normalized proxies. Their numerator and denominator windows are not established as identical.
- The tenure unit is unspecified in the dictionary. Avoid labeling the intensity as orders per month.



E-commerce: cost-sensitive random forest

- A forest averages churn probabilities across decision trees. Each tree learns split rules from a bootstrap sample.
- Balanced class weights increase the influence of the less frequent churn class during fitting.
- Data partition: 3,044 train, 1,015 validation and 1,015 test records.
- Training chooses the model. Validation chooses an action threshold. The test partition assesses the frozen decision rule.



E-commerce: hyperparameter optimization

- GridSearchCV selects by average precision across three stratified training folds.
- Depth controls tree complexity. Leaf size controls the smallest terminal customer group.
- The grid tests four candidates with 200 trees and `max_features = sqrt`.
- Every fold refits preprocessing. CV selects the model; the separate test set estimates final performance.

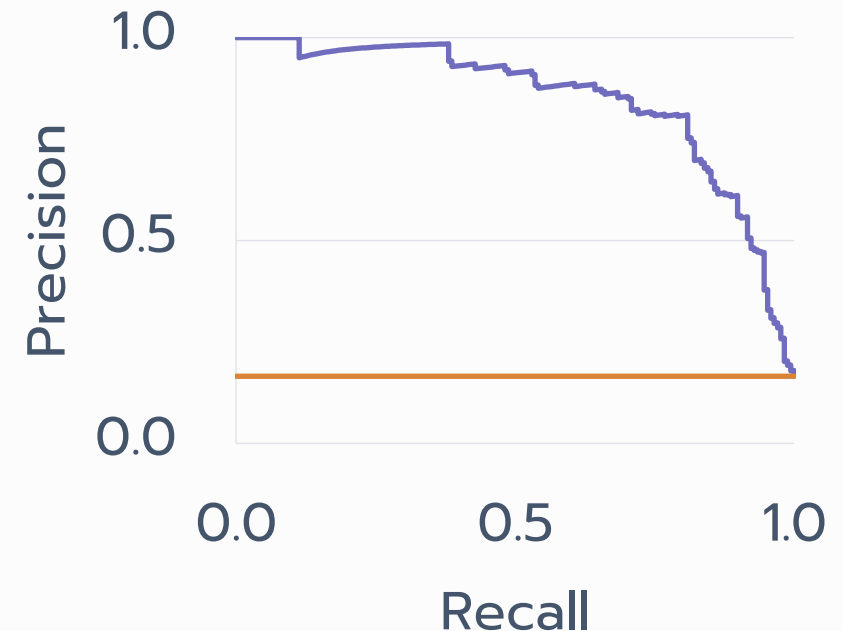
```
grid = {  
    'model__max_depth':  
        [8, None],  
    'model__min_samples_leaf':  
        [2, 5],  
    'model__max_features':  
        ['sqrt'],  
    'model__n_estimators': [200]  
}  
search = GridSearchCV(  
    pipeline, grid, cv=cv,  
    scoring='average_precision',  
    n_jobs=2  
)  
search.fit(X_train, y_train)
```

Source: [scikit-learn, GridSearchCV](#)

E-commerce: ranking performance

- Test PR-AUC (average precision): 0.848.
Test ROC-AUC: 0.945.
- Precision measures how many flagged customers churned. Recall measures how many churners were captured.
- Random-ranking precision is the test churn share, 16.6%.
- Average precision weights precision by recall increments. It differs from trapezoidal PR area.

Precision-recall, held-out test



E-commerce: decision costs

- Validation selected threshold 0.242. Test recall is 89.9% and precision is 58.8%.
- Teaching cost assumption: missed churn costs 5 units; unnecessary contact costs 1 unit.
- Test loss = $5 \times 17 + 106 = 191$ cost units.
- Replace these illustrative costs with verified economics. Detection does not establish that an offer will prevent churn.

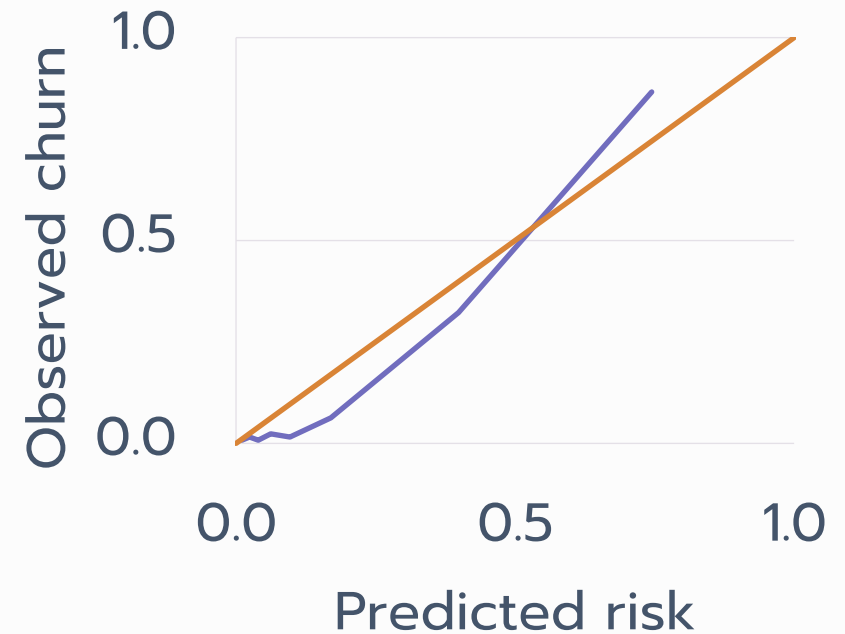
Actual / flag	No	Yes
Retained	741	106
Churned	17	151

False positive: +1 cost; false negative: +5 cost

E-commerce: probability calibration

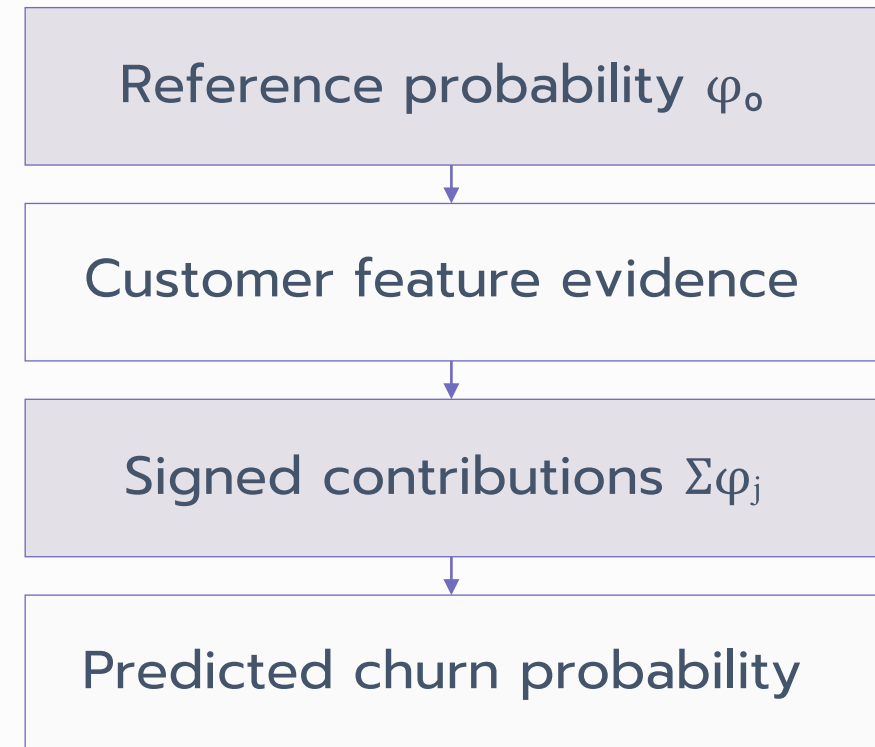
- Test Brier score: 0.0596. Lower values indicate lower mean squared probability error.
- Each bin compares mean predicted risk with observed churn share.
- Above the diagonal means underprediction in that risk region; below means overprediction.
- This is an evaluation. Later recalibration needs training data and fresh holdout validation.

Reliability diagram, test



E-commerce: game-theoretic explanation

- SHAP treats model inputs as players in a cooperative prediction game.
- The explained value is this forest's churn probability for one customer, relative to its training-path reference.
- Each feature receives a signed contribution. The contributions and baseline sum to the prediction.
- A global summary aggregates many local explanations. It describes how the fitted model uses information.



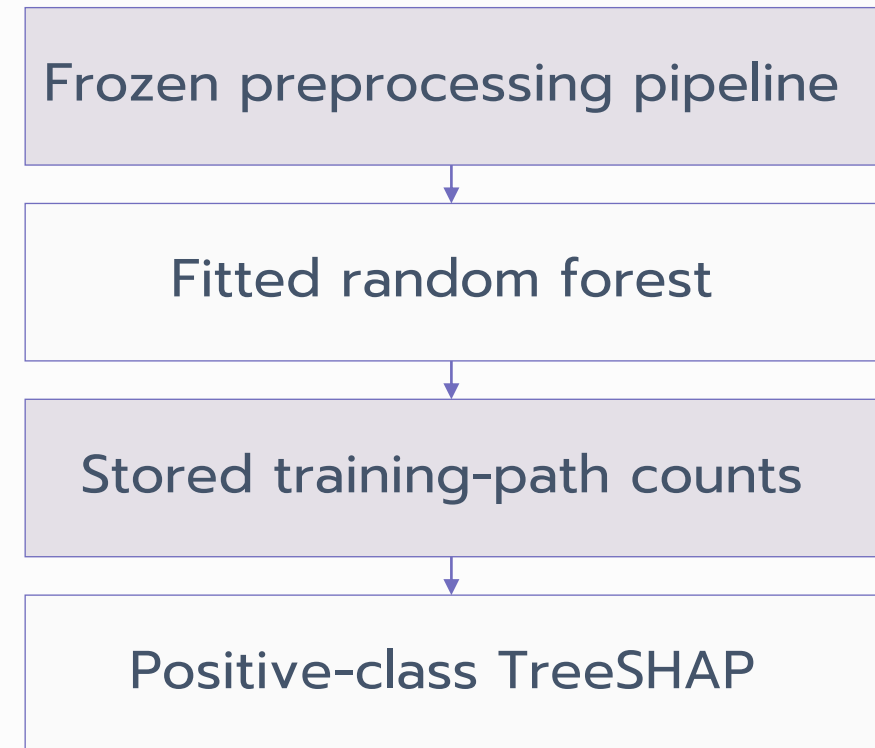
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- Local accuracy requires $f(x) = \varphi_0 + \sum_j \varphi_j$. Positive values raise the model output relative to the baseline.

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Symmetry	Equal role, credit
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E-commerce: TreeSHAP computation

- TreeSHAP exploits tree paths to avoid enumerating all 2^M feature coalitions.
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- KernelSHAP is model-agnostic and samples coalitions. It is useful when a tree-specific engine is unavailable.
- Path-dependent TreeSHAP uses stored training frequencies. Raw output for this forest is class probability.



E-commerce: explanation boundaries

- Correlated inputs and derived ratios can share credit. Rankings depend on the model reference.
- Training-path attribution does not identify causal effects or the true conditional distribution.
- Grouped dummy columns preserve the total. They aggregate the original game for display.
- Positive SHAP means higher model risk. Retention impact requires a controlled study.

Evidence	Interpretation
High risk	Prioritize review
Positive φ	Raises prediction
Mean $ \varphi $	Model reliance
Randomized lift	Action effect

Source: SHAP, TreeExplainer documentation; Lundberg & Lee (2017), SHAP

E-commerce: advanced engine architecture

- Transform test rows with the frozen ColumnTransformer. The model stores the training-path reference.
- For a scikit-learn random forest, model_output='raw' returns class probabilities, including churn class 1.
- Resolve the class axis across SHAP versions before constructing a shap.Explanation.
- Assert that the baseline plus feature contributions reproduces each customer's predicted risk.

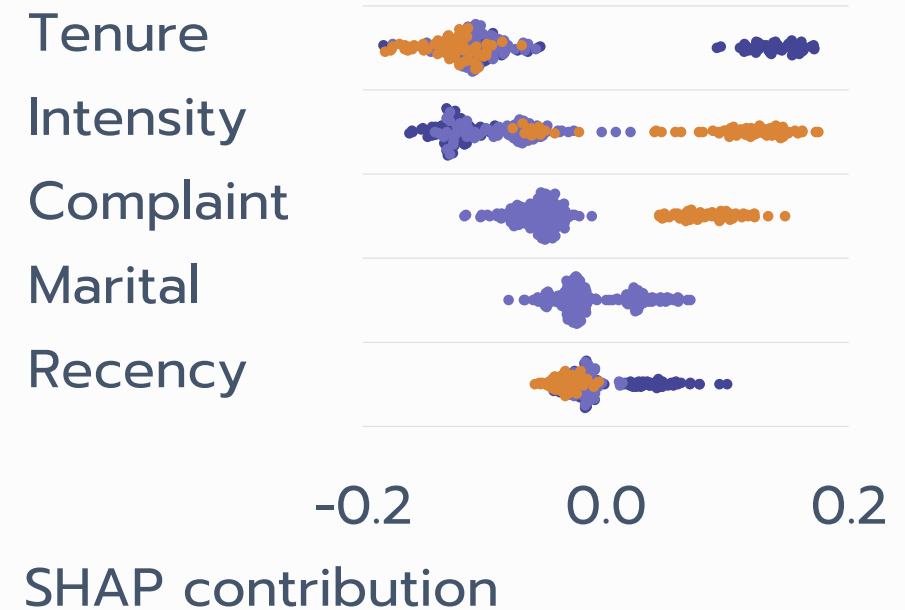
```
engine = shap.TreeExplainer(
    forest, model_output='raw',
    feature_perturbation=
        'tree_path_dependent'
)
raw = engine.shap_values(Z)
phi = raw[:, :, 1]
base = engine.expected_value[1]
np.testing.assert_allclose(
    base + phi.sum(1), p,
    atol=1e-5, rtol=0
)
```

Source: SHAP, TreeExplainer documentation

E-commerce: global SHAP beeswarm

- Largest mean absolute attribution: Tenure, 12.0 percentage points.
- Each dot is one explained test customer. Its horizontal position is that feature's contribution to predicted churn.
- Blue marks lower numeric values; orange marks higher ones. Categorical and missing values use neutral purple.
- Feature order summarizes model reliance. It does not rank the profit from changing those features.

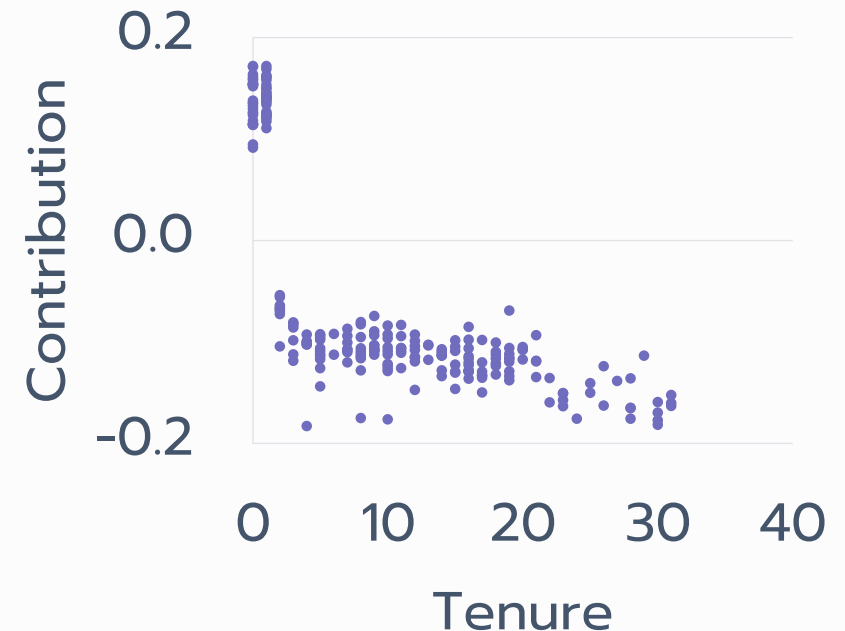
TreeSHAP, top five features



E-commerce: nonlinear attribution pattern

- Horizontal: tenure. Vertical: its grouped contribution to predicted churn.
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- Vertical spread reflects other inputs and interactions. Sparse tails need caution.

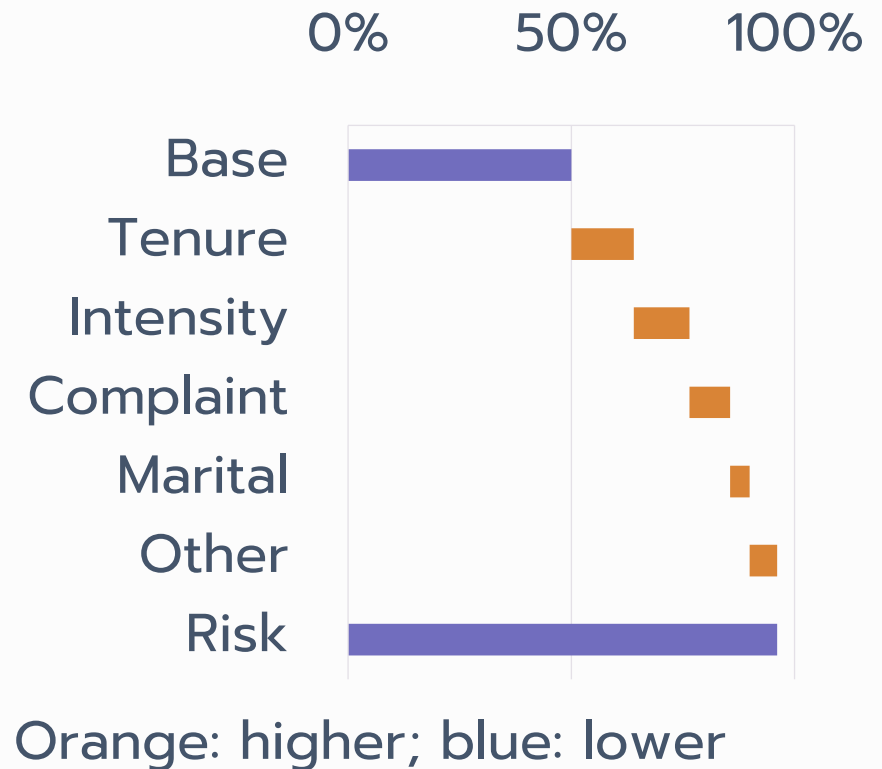
SHAP dependence, test sample



E-commerce: one high-risk customer

- Model reference: 50.0%. This customer's predicted risk: 96.1%.
- The waterfall starts at the reference probability and adds each signed contribution.
- The four largest effects are visible. Other features aggregate the remaining contributions exactly.
- The customer is selected by predicted risk, without using the observed label to choose the example.

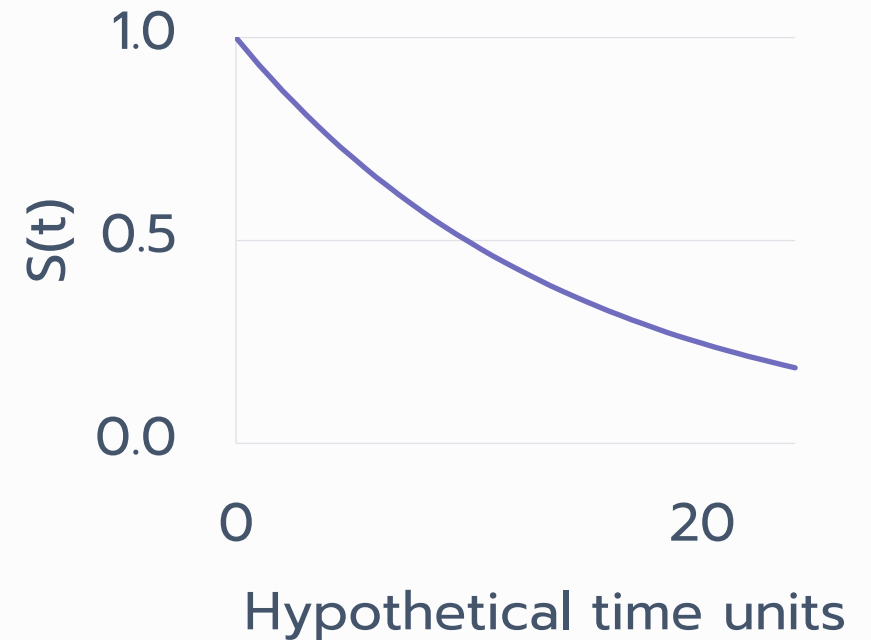
Local probability waterfall



E-commerce: survival and hazard over time

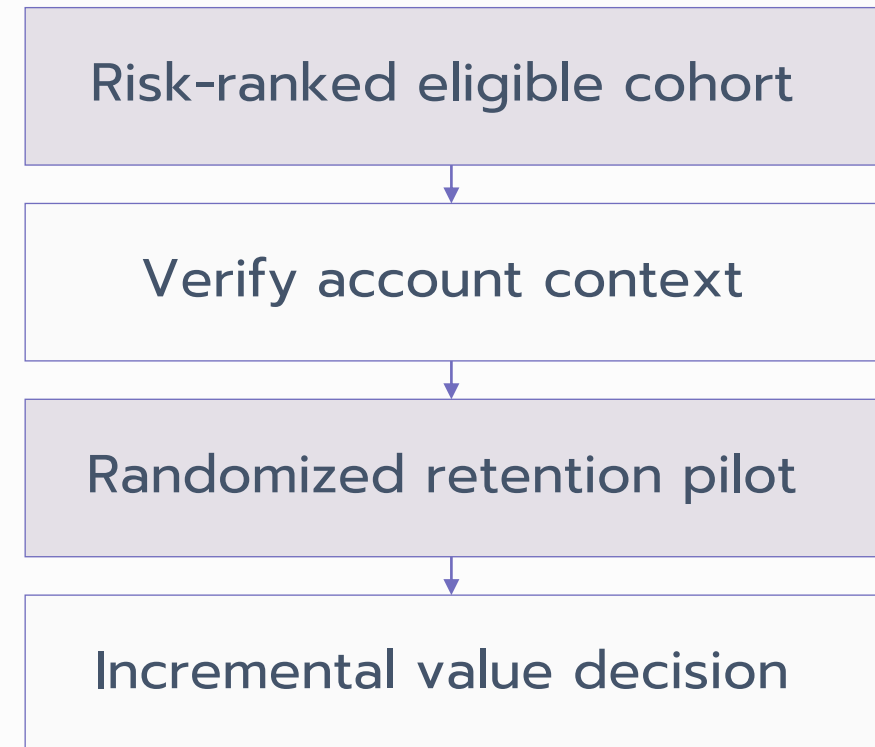
- Survival is $S(t) = P(T > t)$. Hazard is the event rate among customers still at risk.
- Event dates, censoring dates and an observation origin are absent. Empirical survival is not identified.
- Hypothetical only: hazard $\lambda = 0.070$ per time unit; $S(t) = \exp(-\lambda t)$.
- A longitudinal extension needs join date, event or censoring time and features known before each interval.

Hypothetical survival only



E-commerce: executive retention playbook

- Observed evidence: test AP 0.848; the largest model attribution is tenure.
- Priority hypothesis: test onboarding and complaint-resolution support in eligible segments.
- Randomize eligible accounts to treatment and control. Report incremental retained margin after contact and incentive costs.
- Launch gate: verified timing, independent future validation and a positive incremental value estimate. Pause if any gate fails.



Source: Kaggle: ankitverma2010; scikit-learn, Classification metrics; SHAP, TreeExplainer documentation; lecture computation

Customer Churn Analytics

Credit card and e-commerce customers

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